# Problem 4.8

You invest \$1,000 now, at an annual simple interest rate of 6%. What is the effective rate of interest in the fifth year of your investment?

## Solution

For simple interest,

$$a(t) = 1 + it.$$

Here,

$$i = 6\% = 0.06.$$

Thus,

$$a(5) = 1 + 0.06(5) = 1 + \frac{30}{100} = 1.3,$$

and

$$a(4) = 1 + 0.06(4) = 1 + \frac{24}{100} = 1.24.$$

The effective rate of interest in the fifth year is

$$i_5 = \frac{a(5) - a(4)}{a(4)}.$$

Therefore,

$$i_5 = \frac{1.3 - 1.24}{1.24} = 0.048387.$$

Hence,

$$i_5 \approx 4.8387\%.$$